Style
M12 Plug, shielded
8 pin
A-coding

Ratings
Voltage: 30 V
Current: 2 A

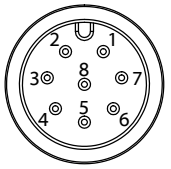
Operating Temperature Range
-40 ~ 105 °C

Materials
① Insulator: Polyamide 66 (PA66), UL94 HB, black
② Pin: C3604 brass, gold plated
③ Shell: C3604 brass, nickel plated
④ Nut: C3604 brass, nickel plated
⑤ O-ring: Silicone rubber
⑥ Pin alignment guide: Polyamide 66 (PA66), UL94 HB, black

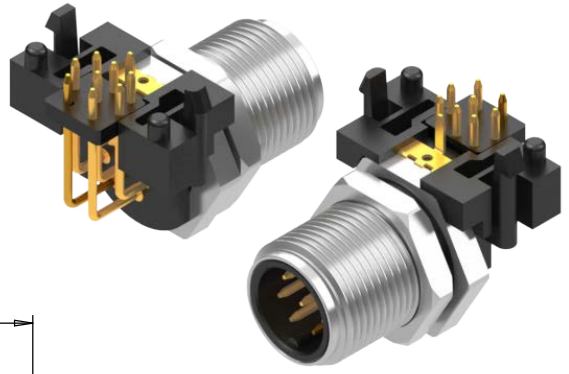
Electrical Requirements
Dielectric strength: 1 min @ 500 Vac
Insulation resistance: 100 MΩ @ 500 Vdc
Contact resistance: 10 mΩ or less

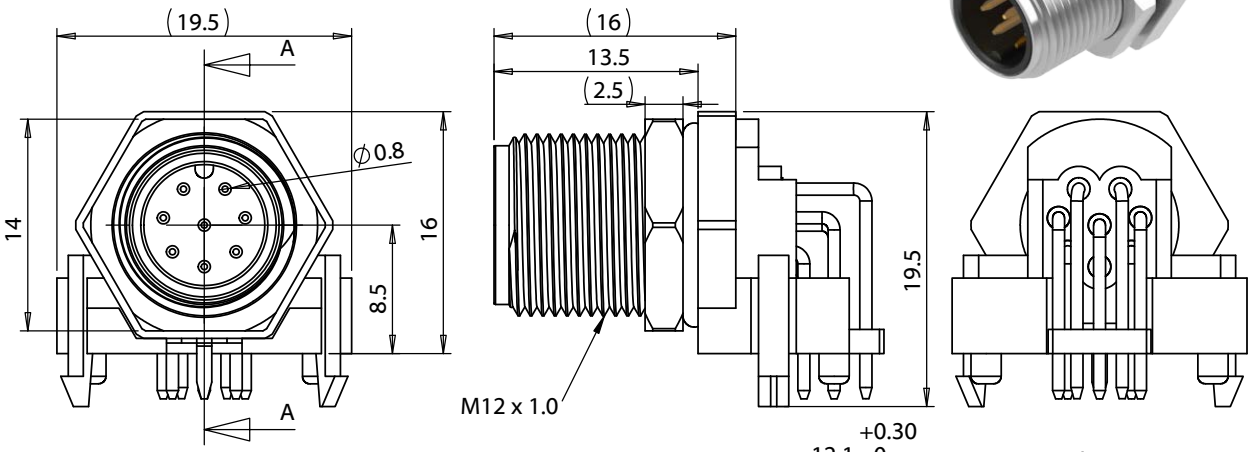
Mechanical Requirements
Life cycle: 500 mating cycles minimum

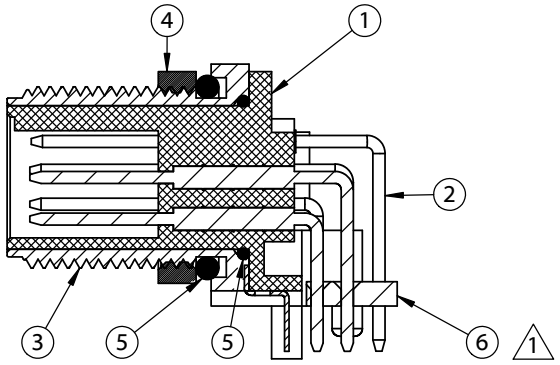
Environmental Requirements
Salt spray test: 35 °C, relative humidity 90-95%, 5% NaCl mist for 48 hrs. Wash parts after test. Maintain mechanical requirements and a contact resistance of less than 10 mΩ.
Waterproof: IP67 (in mated state). IP67 panel ingress requires sealing cap



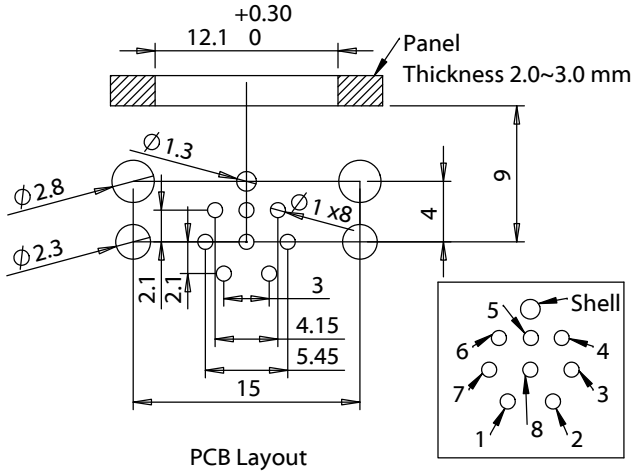
Front View
Not to Scale







SECTION A-A



PCB Layout

Revision:	Date:	Description:
A	9/30/2020	Initial release
A1	10/13/2021	Added pin alignment guide

Prepared: AL Digitally signed by AL Date: 2025.02.28 15:28:43 -08'00'

Verified: RR Digitally signed by RR Date: 2025.03.03 06:59:45 -08'00'

Dimensions are in millimeters.
Tolerances:
X: ± 0.5 mm
X.X: ± 0.3 mm
X.XX: ± 0.1 mm

Notes:
RoHS compliant

Function test: no open, no short circuit, no intermittent

Description:
Connector, M12 A-code plug, 8 pin, 12xL24.9 mm, panel mount, thru hole, 90°, board lock, gold plated, IP67

TENSILITY
tel 1.541.323.3228 800 877.670.7118
fax 1.541.323.4202 web tensility.com

Size: A
Part number: 54-00251
Scale: 2:1
Sheet 1 of 1

5 4 3 2 1